- These layers apply a set of filters (or kernels) to the input image, producing feature maps. Each filter is designed to detect specific patterns, such as edges, textures, or colors.
- The convolution operation preserves the spatial relationship between pixels by learning image features in small local regions.
- Example parameters include the number of filters, filter size (e.g., 3x3), stride, and padding.

2. **Pooling Layers**:
    - Pooling layers reduce the spatial dimensions of the feature maps, decreasing the computational requirements and helping to prevent overfitting.
    - Common pooling methods include MaxPooling (selects the maximum value in a region) and AveragePooling (computes the average value).
    - Pooling helps retain the most important features while discarding redundant information.
3. **Fully Connected Layers**:
    - These layers are traditional dense layers that connect every neuron in one layer to every neuron in the next.
    - Fully connected layers aggregate the spatially extracted features from convolutional and pooling layers to make final predictions.
4. **Activation Functions**:
    - Non-linear activation functions such as ReLU (Rectified Linear Unit) introduce non-linearity, enabling the model to learn complex mappings.
    - The final layer typically uses a softmax activation function for multi-class classification tasks, producing probabilities for each class.
5. **Dropout Layers** (Optional):